

SPLADE – Sparse Lexical and Expansion Model for First Stage Ranking (2021)

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2022.10.09

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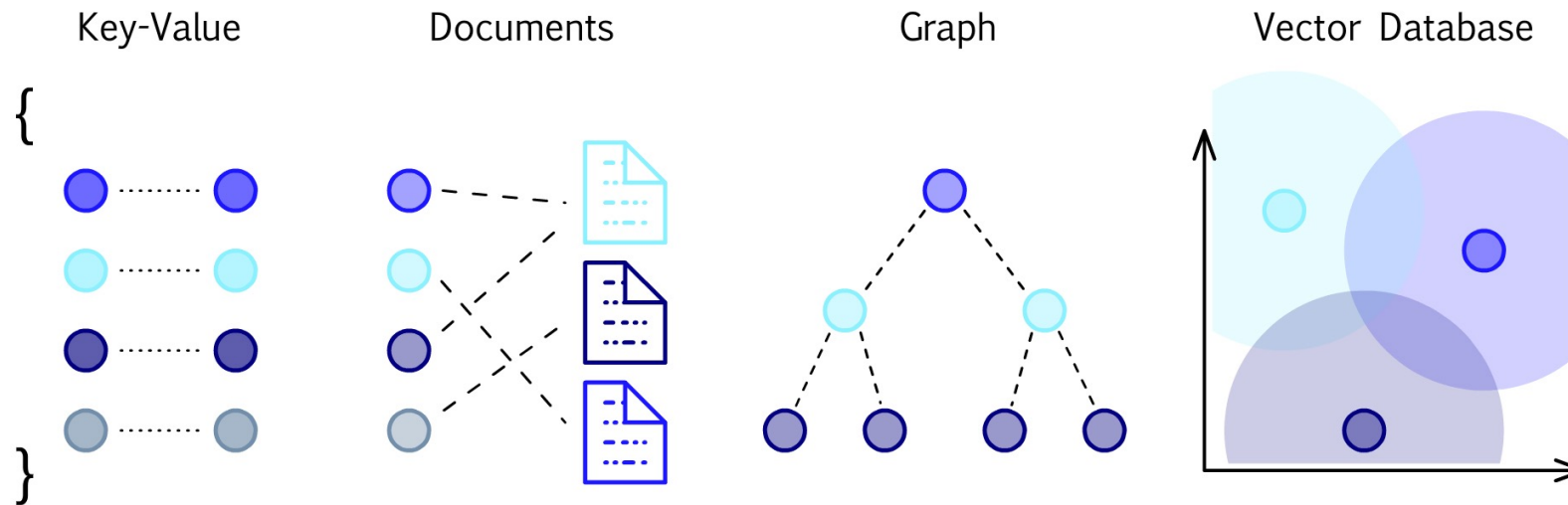
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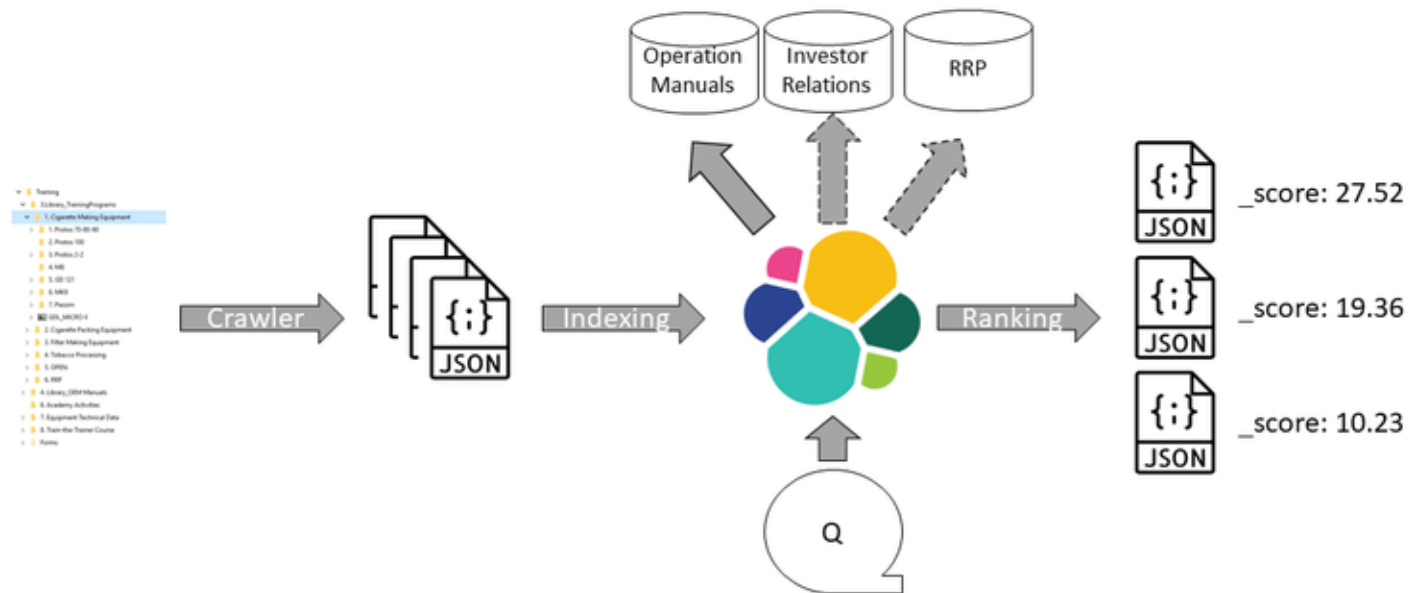
Retriever models

- Answering Questions System
 - **Vector database** : Storing information-rich vectors that numerically represent the meaning of contexts.



Retriever models

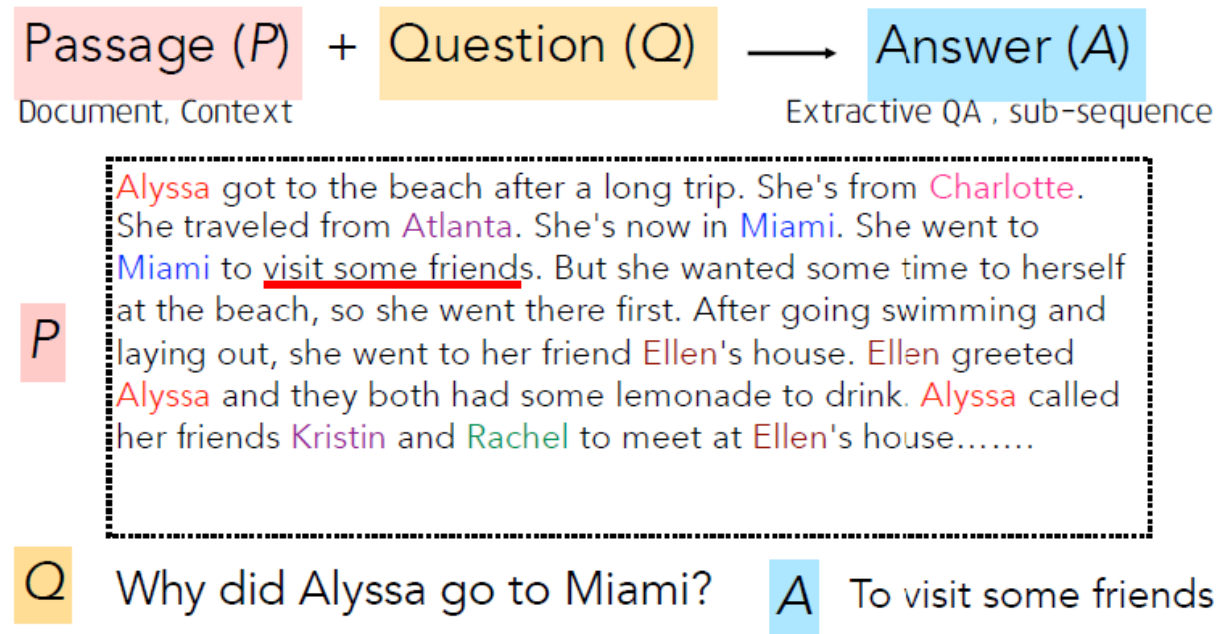
- Answering Questions System
 - **Retriever model** : Encoding questions(**query**) and contexts(**passage**) into the same vector space



Retriever models

- Answering Questions System

- **Reader model** : Taking questions and context and attempts to identify a span from the context which answers the question.



Stages of search : Ranking and reranking

- Ranker
 - Pre-selecting candidates set of documents from billions of text
 - Term-based-model : bm25, inverted index
 - Allowing quick access to the documents that contain the query terms
 - Search engine requires detecting and counting words – sparse words (near zero values)
- Re-ranker
 - Finding an order of decent document for query from candidate set
 - More computations than ranker in general

Understanding NLP for information retrieval

- PLM Model
 - BERT-IR
 - ranking model have SOTA by a large margin
 - Sentences embeded as hundred-dimensional continuous vectors
 - Backbone of efficient architectures for first-stag ranking
- Sparse Model
 - TF-IDF, BM25, LSI(Lexical Semantic indexing)
- BERT
 - ANN(Approximate nearest neighbouring) : Repositiong vectors and creating index of each document by BERT

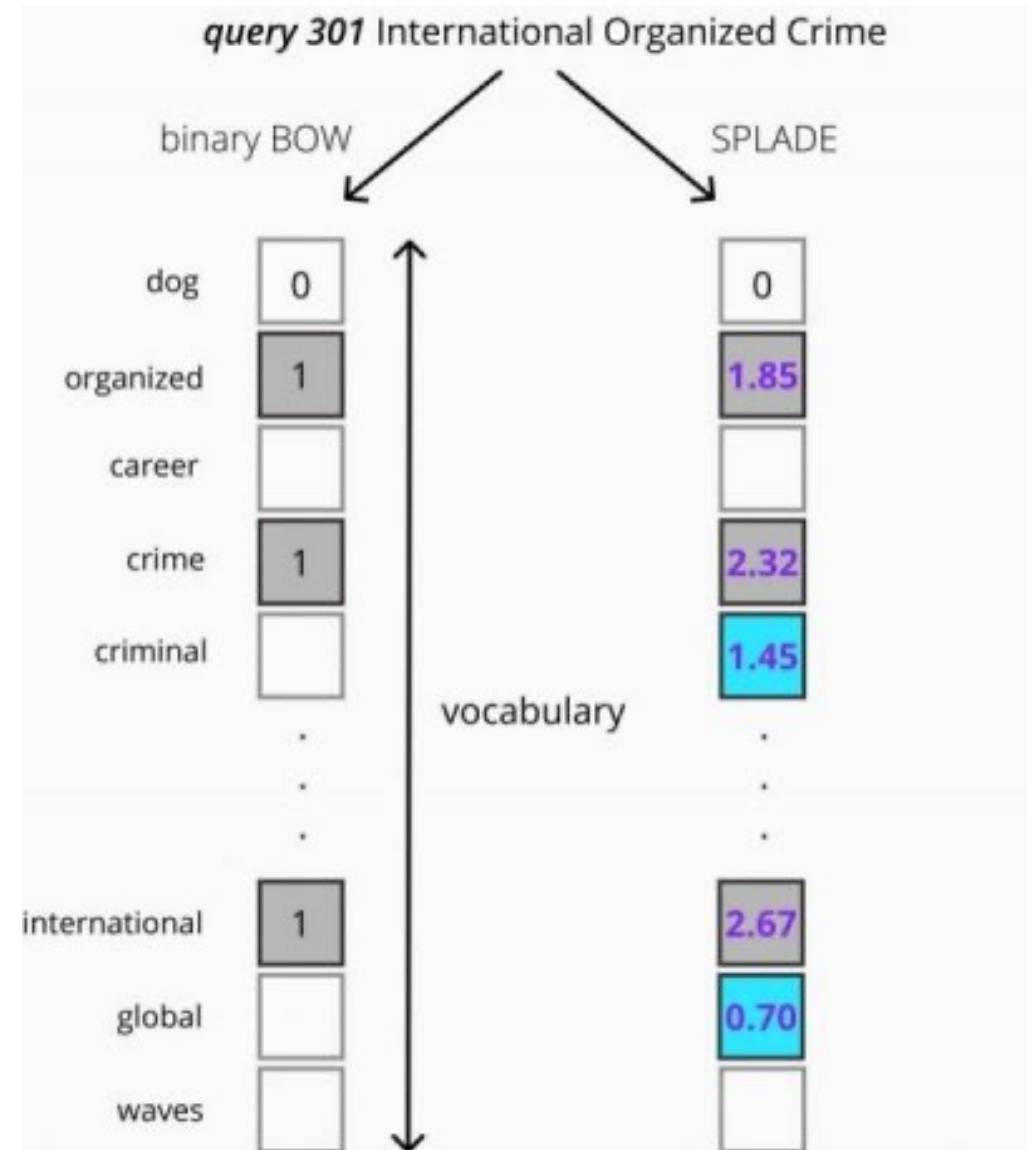
Approaches to indexing : Dense vs Sparse

- **Sparse Representation (Lexical)**
 - Corresponding actual specific words(or subword)
 - Easier to interpret how documents are ranked by a given query (pros)
 - Low cost for building new sparse search engine infrastructure (pros)
 - Hardly analysis of blackbox of BERT (cons)
- **Dense Representation (Semantic)**
 - Contextualized representation
 - Outperform for datasets than sparse representation learning in general

Sparse Lexical AnD Expansion (SPLADE)

Aim: learn efficient sparse representation for first-stage retrieval

- Adding new terms
- Removing existing terms
- Estimating a term importance



Sparse Lexical AnD Expansion (SPLADE)

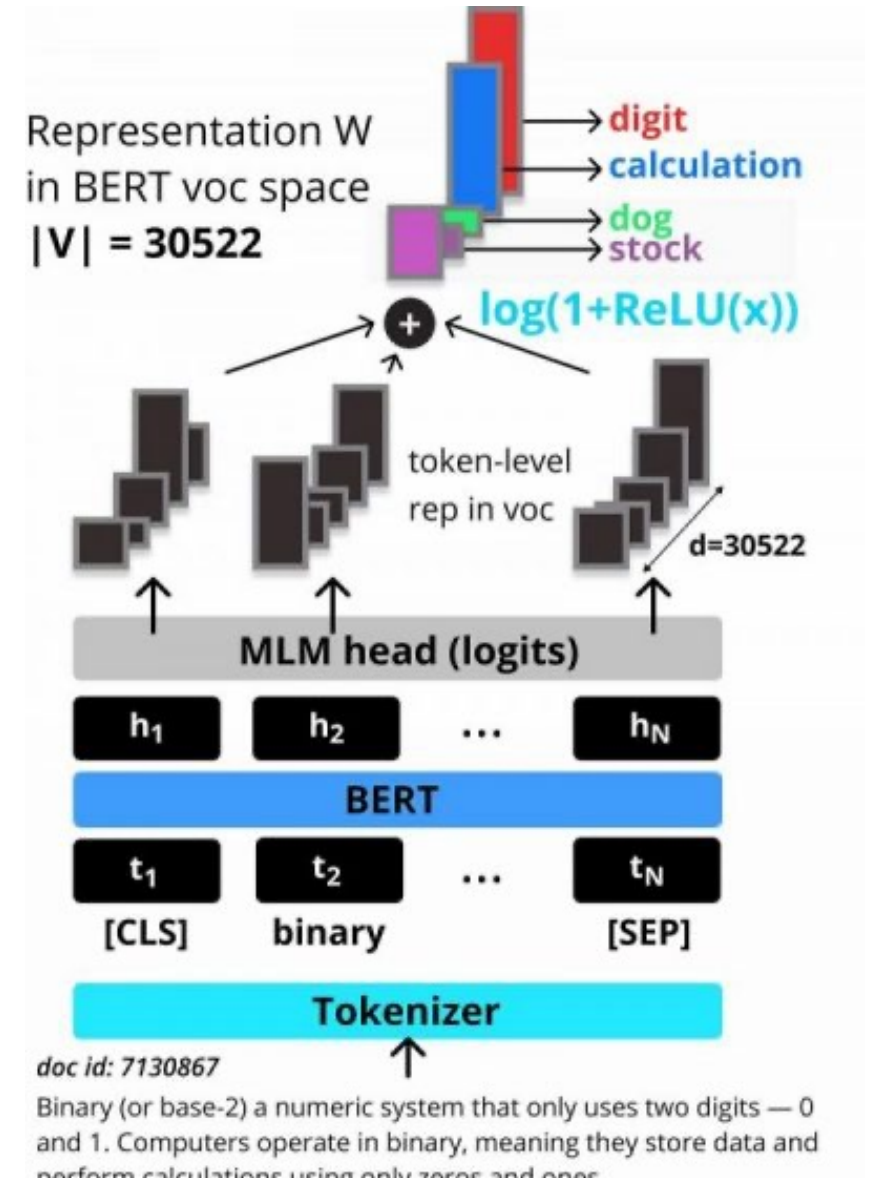
some terms are dropped (compression effect)

original document (doc ID: 7131647)
if (1.2) bow (2.56) legs (1.18) is caused (1.29) by (0.47) the bone (1.2) alignment (1.88) issue (0.87) than you may be able (0.29) to correct (1.37) through (0.43) bow legs correction (1.05) exercises. read more here.. if bow legs is caused by the bone alignment issue than you may be able to correct through bow legs correction exercises.
expansion terms / stemming effect good expansion terms
(leg, 1.62) (arrow, 0.7) (exercise, 0.64) (bones, 0.63) (problem, 0.41) (treatment, 0.35) (happen, 0.29) (create, 0.22) (can, 0.14) (worse, 0.14) (effect, 0.08) (teeth, 0.06) (remove, 0.03)
bad expansion terms !

- Model reweighting each term by emphasizing important terms
- Model discarding most of those non-information content
- Expansion rich document
- Adding stemming effects (legs -> leg)

Sparse Lexical AnD Expansion (SPLADE)

- Tokenization
- Embedding each (sub)token to project back the full of vocabulary
- Summing up tokens and apply to activation layers
- Saturating into single-terms (by one document or query)
- Rank the score query and document by dot product

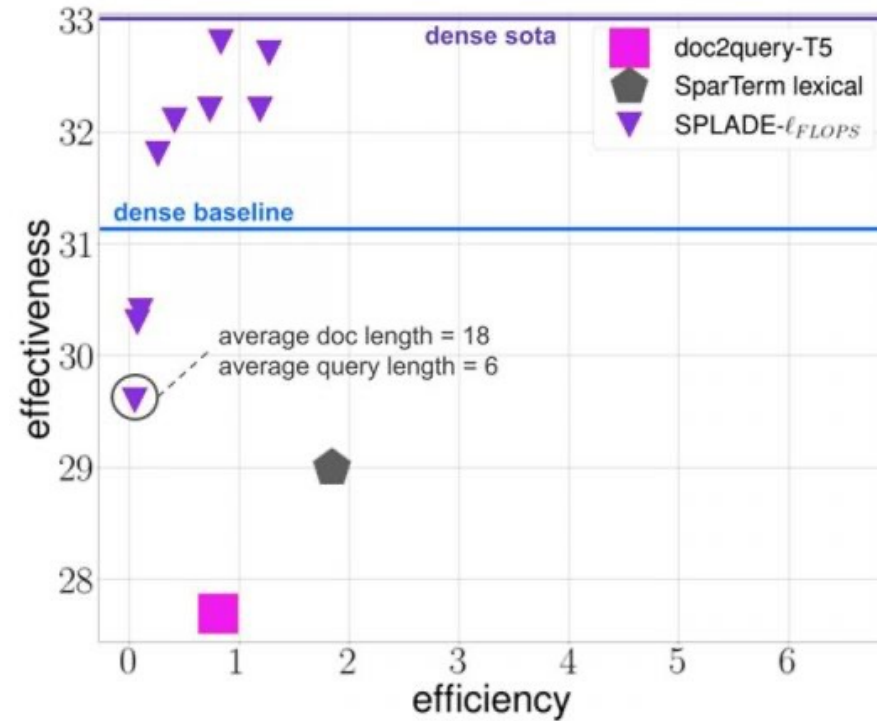


Sparse Lexical AnD Expansion (SPLADE)

Query	SPLADE BOW rep	New terms (expansion)	Dropped terms (compression)
what is paranoid sc	(paranoid, 2.71), (carolina, 1.88), (sc, 1.85), (paranoia, 1.68), (suspicion, 0.4), (suspect, 0.18), (fear, 0.17), (security, 0.09)	security, suspect, carolina, paranoia, suspicion, fear	what, is
what is prime rate in canada	(prime, 2.62), (canada, 2.47), (rate, 2.12), (canadian, 1.58), (rates, 0.15)	rates, canadian	in, is, what
who plays young dr mallard on ncis	(mall, 2.48), (nc, 2.05), (dr, 2.01), (##ard, 1.91), (young, 1.8), (played, 1.49), (carolina, 1.09), (doctor, 1.0), (actor, 0.56), (##is, 0.33), (younger, 0.07), (little, 0.05)	younger, doctor, carolina, little, actor, played	plays, on, who
what is milk thistle useful in treating?	(thistle, 2.77), (milk, 2.35), (useful, 1.04), (used, 0.75), (treat, 0.74), (treatment, 0.63), (medication, 0.5)	treatment, medication, treat, used	?, what, in, is, treating
what is marilyn manson music called	(manson, 2.59), (music, 2.26), (marilyn, 2.13), (called, 1.32), (tate, 0.75)	tate	is, what
Does Suddenlink Carry ESPN3	(espn, 2.29), (sudden, 2.16), (##link, 1.9), (carry, 1.55), (##3, 1.53), (olympic, 1.2), (link, 1.11), (broadcast, 0.91), (carrying, 0.37)	carrying, olympic, broadcast, link	does

Conclusion

- Trade-off : Performance and Efficiency
- Adjusting ranking loss and regularization loss
- Comparing with DPR (without FLOPS)



References

- <https://europe.naverlabs.com/blog/splade-a-sparse-bi-encoder-bert-based-model-achieves-effective-and-efficient-first-stage-ranking/>
- <https://www.pinecone.io/learn/retriever-models/>
- <https://www.pinecone.io/learn/vector-database/>

Thank you

